

INCLINED PLANES – FURTHER PROBLEMS

09/01/26

1. The combined mass of a cyclist and her bicycle is 60 kg. This cyclist is travelling at a speed of 10 ms^{-1} when she starts to free-wheel down a hill of inclination 1 in 20. If there are constant resistances of magnitude 20 N acting on her during this motion, find her speed after she has travelled 200 m down the hill.

2. A BMX bike is being ridden up a smooth ramp inclined at 20° to the horizontal. The base of the ramp is at ground level.

The total mass of the BMX and rider is 80 kg, and the driving force on the bike is 1200 N.

The BMX is initially travelling at 5 ms^{-1} and the ramp is 6 m long.

- Find the speed of the BMX at the end of the ramp.
- Find the height above the ground

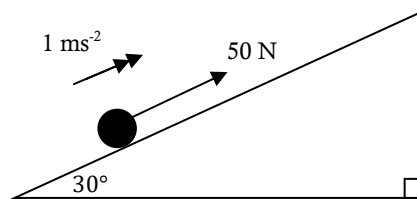
Once it reaches the end of the ramp, the BMX travels through the air as a projectile.

- Find the horizontal range of the bike.

3. A crate of mass 100 kg being raised up a rough plane inclined at 5° to the horizontal by means of a cable which runs parallel to the line of greatest slope. When the tension in this cable has magnitude 200 N the crate travels at constant velocity.

Find the frictional force (assumed constant) acting on the crate.

4. A particle of mass 8 kg is at rest on a rough plane inclined at 30° to the horizontal. When a force of 50 N is applied directly up the plane the particle accelerates up the plane at a rate of 1 ms^{-2} .



- Find the magnitude of the frictional force acting on the particle.

The 50 N force is now removed, and replaced with a 50 N force acting directly *down* the plane. A frictional force of equal magnitude to that found in Part a now acts directly up the plane.

- Find the new acceleration of the particle.

5. From M1 January 1998:

A cyclist starts from rest and moves along a straight, horizontal track. The forward driving force is a constant 72 N and the resistances to forward motion are a constant 20 N. The combined mass of the bicycle and cyclist is 80 kg.

- Calculate the acceleration of the cyclist. [4]

- How long does it take the cyclist to reach a speed of 8 m s^{-1} ? How far does the cyclist travel in this time? [4]

The cyclist starts to climb a hill when her speed reaches 8 m s^{-1} . The hill is at a constant angle of 10° to the horizontal. The driving force and resistances to motion are unchanged.

- Show that the cyclist is now decelerating at about 1.05 m s^{-2} . [4]

The cyclist cannot ride at less than 1 m s^{-1} up the hill and she gets off when she has slowed down to that speed.

- Calculate the total time elapsed and the total distance travelled from the start to the end of the ride. [3]